

IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Electric Vehicle Modeling and Simulation Design

Virtual Internship

AICTE – EduSkills Virtual Internship Program
Supported by MathWorks

shivam kumar | B.Tech (Computer Science and Engineering)

June – August 2026

ABOUT THE STUDENT

Candidate Profile

Student Name	shivam kumar
Stream	B.Tech
Branch	B.Tech (Computer Science and Engineering)
Institute	IILM University, Greater Noida
Internship Domain	Electric Vehicle Modeling and Simulation Virtual Internship Program
Duration	8 Weeks June – August 2026
AICTE Student ID	STU699ac8fcc6e891771751676

ABOUT THE PROGRAM

AICTE – EduSkills Virtual Internship Program

1

8-Week Program

Selected for an 8-week virtual internship with a structured, week-wise learning plan.

2

Learning + Practice

The program combines structured learning with practical exposure and project documentation.

3

Assessment

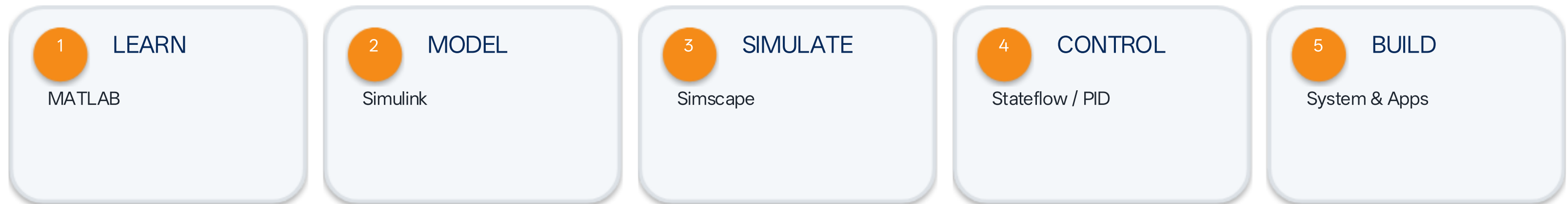
Weekly assessments, project submissions and a final assessment contribute to the final grade.

Successful completion of the required steps and assessments leads to a Virtual Internship Certificate.

LEARNING FOCUS

From MATLAB Fundamentals to EV-Oriented System Modeling

The internship progresses from MATLAB and Simulink fundamentals to control design, physical-system simulation, battery management, power systems, system architecture and application development.



PROJECT OBJECTIVES

What the Internship Set Out to Achieve

1 MATLAB Fundamentals

Learn MATLAB®, its language, development environment and essential engineering-data tools.

2 Simulink Modeling

Create, edit and simulate block-diagram models representing real-world systems and algorithms.

3 Control Design

Learn feedback control design in Simulink®, including PID controller tuning.

4 Physical Simulation

Use Simscape® for physical systems, circuit simulation and power-electronics converters.

5 Battery & Power Systems

Explore battery management, thermal management, charging/discharging and microgrid simulation.

6 System & App Development

Learn system architecture with System Composer™ and build applications with App Designer.

TECHNOLOGIES & TOOLS

The Toolkit Behind the Internship

1 MATLAB®

Programming, analysis and engineering data

2 Simulink®

Block-diagram modeling and simulation

3 Simscape®

Physical-system modeling and simulation

4 Stateflow®

Supervisory control and decision logic

5 Simscape Battery

Battery modeling, BMS and thermal management

6 Simscape Electrical™

Circuits, converters and power systems

7 System Composer™

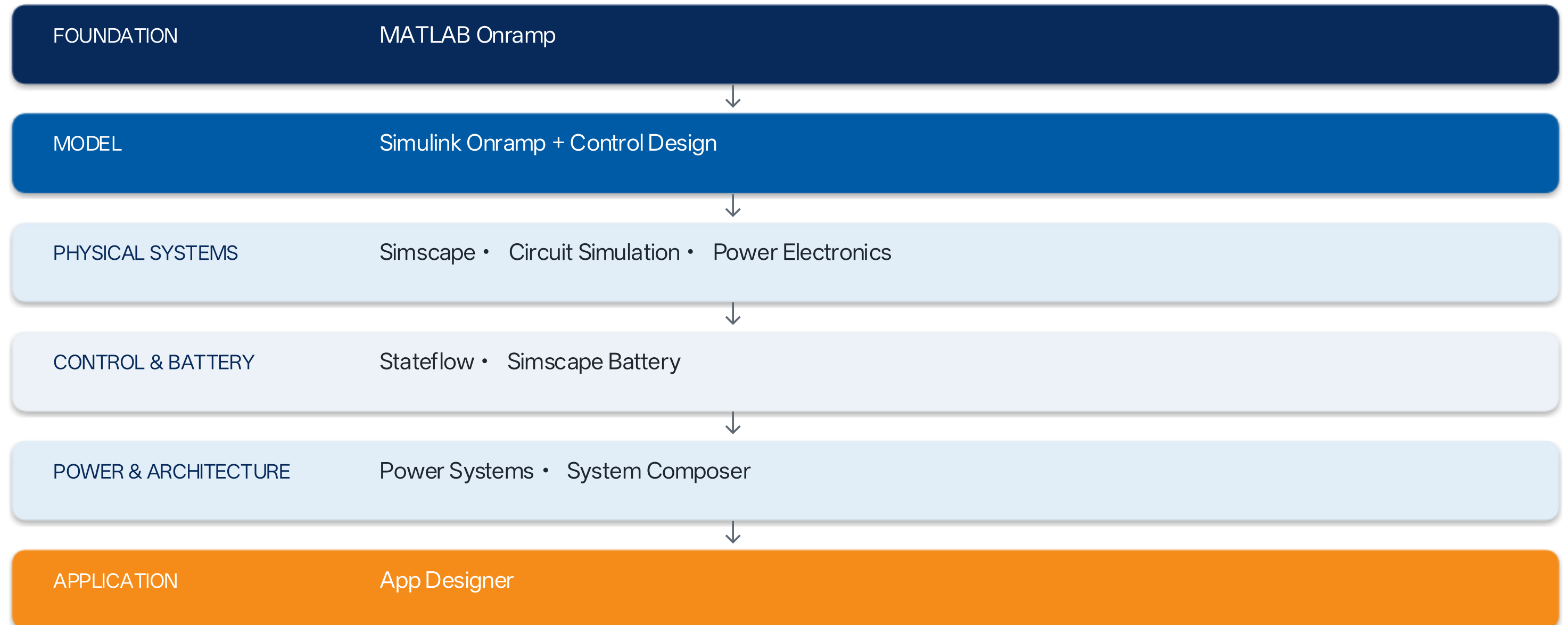
System architecture and requirements tracing

8 App Designer

User interfaces, callbacks and applications

SYSTEM ARCHITECTURE

Learning Pipeline: From Modeling to Application



METHODOLOGY

The 8-Week Learning Journey

1

WEEK 1 — MATLAB Onramp

MATLAB language, development environment and tools for analyzing science and engineering data.

2

WEEK 2 — Simulink + Control Design

Block-diagram modeling, simulation and feedback control with PID tuning.

3

WEEK 3 — Simscape + Circuits + Power Electronics

Physical networks, electrical circuits and buck-converter simulation.

4

WEEK 4 — Stateflow Onramp

Supervisory control systems and decision logic using state machines and flow charts.

5

WEEK 5 — Simscape Battery Onramp

Battery modeling, BMS, thermal management and charging/discharging algorithms.

6

WEEK 6 — Power Systems Simulation

Microgrid modeling, Simscape Electrical and power-component control strategies.

7

WEEK 7 — System Composer Onramp

System architecture, requirements tracing and Simulink integration for verification.

8

WEEK 8 — App Building Onramp

MATLAB App Designer, user-interface design, callbacks and complete app development.

EXPECTED OUTCOMES

Skills & Capabilities Gained

- 1 Ability to use MATLAB for engineering-oriented analysis and learning.
- 2 Practical understanding of block-diagram modeling and simulation in Simulink.
- 3 Exposure to physical-system, circuit and power-electronics simulation with Simscape.
- 4 Understanding of supervisory control and decision logic using Stateflow.
- 5 Hands-on understanding of battery modeling, BMS and thermal-management concepts.
- 6 Experience with microgrid and electrical power-system simulation.
- 7 Understanding of system architecture and requirements tracing using System Composer.
- 8 Capability to build MATLAB applications using App Designer.

CERTIFICATION

Successfully Completed

Grade: O (Outstanding)

Certificate ID: 4ad8bd95e2683dcd3be1

Student ID: STU699ac8fcc6e891771751676

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शिक्षा मंत्रालय
MINISTRY OF
EDUCATION

शिवम कुमार

2026
VIRTUAL INTERNSHIP

EduSkills®
Nation Building Through Skills

**NATIONAL
INTERNSHIP
PORTAL**

Certificate of Virtual Internship

This is to certify that

shivam kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Electric Vehicle Modeling and Simulation Design Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By **MathWorks®**

Chandan Pramanik
Director- Education
MathWorks India

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

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GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

THANK YOU

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